

Music-Gen v3 RULES Milestone — Fanout Clone 2: First Activation of M-V3-RULES-1 (Cycles 1-2)

2026-09-02

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Abstract

This report covers Cycles 1 and 2 of a fanout-clone branch spawned from the Music-Gen v3 campaign to open the M-V3-RULES-1 milestone for the first time. The clone (fork d5530f8d1ccc, clone 2) was assigned under the c23 operator directive’s “determinism-for-generation-half” stance and the operator’s 2026-09-02 “KEEP MOVING, DO NOT WAIT FOR APPROVAL” directive that closed M-V3-FOCUS-1 with three-of-three operator-ear accepts on the mandatory Chicken Grease and two focus songs (What If I Go and Disco A). The clone’s scoped objective was to design and prototype a deterministic rules-extraction program over the operator-approved v3-rendered corpus — the four focus-song merged.mid deliveries from Cycles 5, 20, and 21 (Chicken Grease 31a164f845f8e27e, What If I Go 252eb21ce7df7328, Rome 51e433ade2a845e1, Disco A cdd2717e52820ff6) — with a frozen three-way rubric_hash_v3_rules byte-equality chain committed before any script was written and a byte-determinism x2 gate on the emitted rules artifact. Cycle 1 delivered the required output artifact docs/v3_rules_deterministic_extractor_spec_c23.md (13 058 bytes, SHA e81ff589200f6d6b52d7a68f813e8120a544bfb371f8e0a45a2ac7f8793d71e5), landed the prototype extractor at scripts/v3_rules/extract_rules.py, and emitted the rules artifact at data/v3/rules/rules_artifact.jsonl (47 662 bytes, SHA e19fb205b282dabbf9f6ba38d97ed53649d160a9bf36e9588e03b7cd71ac8186) — 76 typed rules across five rule types (harmonic 18, rhythmic 18, melodic 18, form 18, arrangement 4) with per-stem provenance from the v3 doctrine — plus 15/15 tests green, 33 read-only anchors preserved byte-identical, a fetchability ladder recording the survey-open-source-first library enumeration, and an internal auditor verdict V3_RULES_LANDS_pending_operator with operator_ear_dependency=false because M-V3-RULES-1 has no ear-gate. Cycle 2 was an explicit bookkeeping-only no-op cycle per the research brief; the auditor performed live SHA re-verification on every anchor with byte-exact match to the Cycle 1 delivery and

closed the branch under COMPLETE and [[BRANCH_COMPLETE]]. This is the first activation of the operator’s rules-hashed contract for the M-V3 arc, and it opens the campaign’s generation-half by materializing a corpus of extracted rules that a subsequent M-V3-GEN-1 milestone can consume as its input.

1. Introduction and scope

The v3 campaign’s milestone structure required M-V3-SPINE-1 (per-song per-stem reconstruction pipeline) and M-V3-FOCUS-1 (at least three focus-song accepts) to close before the downstream milestones — M-V3-CORPUS-1 (corpus breadth), M-V3-RULES-1 (rules extraction), M-V3-EAR-1 (ear model), and M-V3-GEN-1 (generation) — could open. M-V3-SPINE-1 closed at Cycle 20 with the operator’s ear judgment on the Cycle 5 Chicken Grease reconstruction. M-V3-FOCUS-1 closed with redundancy across Cycles 20–21 fanout arcs and became fully operator-ear-blessed on 2026-09-02 (three-of-three accepts on Chicken Grease + WIG + Disco A). The operator’s directive on the same date — “KEEP MOVING, DO NOT WAIT FOR APPROVAL” — released the campaign to open the generation-half milestones without waiting for further per-song ear judgment on the remaining PARTIAL cases (Peach Dream, Rome).

The c23 operator directive named the immediate opening as M-V3-RULES-1 under a “determinism-for-generation-half” stance: the rules-extraction step that feeds the eventual M-V3-GEN-1 generator must be byte-deterministic under fixed environment pins so that the generator’s inputs are reproducible cycle over cycle. The directive also invoked the standing “survey-open-source-first” rule (enumerate candidate library dependencies before writing custom logic, with an on-disk-vs-blocked fetchability probe under the proxy that never attempts actual fetches).

This report is the merge-disposition summary for clone 2 of fork d5530f8d1ccc. Sibling clones in the same fork run parallel opening work on other downstream milestones; they are covered in separate reports.

The clone’s scoped objective as issued:

- **Pin the rubric document** docs/v3_rules_deterministic_extractor_spec_c23.md before any script under scripts/v3_rules/, with a three-way rubric_hash_v3_rules byte-equality chain (doc SHA == data/v3/rules/rubric_hash.txt content == verdict field).
- **Enumerate constraint/grammar/statistical-sampler libraries** (music21, mingus, jsonschema-driven grammar, sklearn) via a fetchability probe under the proxy — no fetch attempts, fetchability_ladder.jsonl records what is on disk versus blocked.
- **Prototype scripts/v3_rules/extract_rules.py** that reads the four operator-approved deliveries read-only and extracts typed rules per the c9 rule schema (harmonic, rhythmic, melodic, form, arrangement) but with per-stem provenance from v3 doctrine.
- **Emit** data/v3/rules/rules_artifact.jsonl plus a self-anchor rules_artifact.sha256.
- **Prove byte-determinism x2**: two fresh tempfile.mkdtemp() runs produce byte-identical rules_artifact.jsonl.
- **Preserve read-only predecessor ledgers** (c9 data/rules/ledger.jsonl, c15 ledger_i3_dminor.jsonl, c40 ledger_rated_corpus.jsonl) unchanged; write only to the fresh data/v3/rules/ shard.
- **Hygiene**: interpreter guard /usr/bin/python3, zero PRNG imports (or explicit random_state=0 recorded in the artifact), zero sidecar_nonfactor imports, zero VST3 state APIs.
- **Test suite** at tests/test_v3_rules_deterministic_extractor.py with at least 12 of 15 tests passing.
- **Six named plus two housekeeping ledger events** under M-V3-RULES-1/*.

The required output artifact is docs/v3_rules_deterministic_extractor_spec_c23.md.

2. Cycle 1: full M-V3-RULES-1 opening

2.1 Rubric freeze (before any script)

docs/v3_rules_deterministic_extractor_spec_c23.md (13 058 bytes, SHA e81ff589200f6d6b52d7a68f813e8120a544bfb37) was committed before any script under scripts/v3_rules/, with the mtime hard-verified. Its pinned hash file at data/v3/rules/rubric_hash.txt carries the same SHA verbatim. The rubric defines the extractor's five rule types (harmonic, rhythmic, melodic, form, arrangement), the per-stem provenance schema extending the c9 rule schema with the v3 doctrine's stem-level annotations, the byte-determinism $\times 2$ mandatory sub-clause, the fetchability-ladder survey requirements, and the acceptance rule that panel measurements are never a LANDS gate (FD-6 continues to hold on M-V3-RULES-1 even though the milestone has no ear-gate).

2.2 Fetchability ladder (survey-open-source-first)

The clone probed the on-disk availability of four candidate library families named by the directive without attempting any network fetch, and recorded the outcomes at data/v3/rules/fetchability_ladder.jsonl (484 bytes). The chosen implementation strategy per the ladder outcomes: pure stdlib plus mido (already vendored via the campaign's mido 1.3.3 pin), no music21, no mingus, no jsonschema-driven grammar, no sklearn. This decision minimizes external dependency surface while satisfying the deterministic-extraction contract.

2.3 Prototype extractor and rules artifact

scripts/v3_rules/extract_rules.py reads the four operator-approved deliveries as read-only inputs:

- Chicken Grease (31a164f845f8e27e) — merged.mid from Cycle 5, operator-blessed on 2026-09-02.
- What If I Go (252eb21ce7df7328) — merged.mid from Cycle 21 clone-1 restart (SHA a93f5c2ae16e5cace42b98886f6ce3eae4bb47393bef9d2abe631aadbe526578).
- Rome (51e433ade2a845e1) — merged.mid from Cycle 20 clone-1 (SHA c28b8686684fddfc841a27e96e299a93f1099).
- Disco A (cdd2717e52820ff6) — merged.mid from Cycle 21 clone-0 (SHA-16 7e6f131f07f0d33c).

Determinism mechanism: pure stdlib plus mido; canonical JSON serialization via `_canonical_json(sort_keys=True, separators=(",", ":"))`; fixed extraction timestamp 2026-09-02T00:00:00Z; content-hashed `rule_id` per rule (16-hex derived from the canonical parameters); explicit `parameters_random_state=0` recorded per rule; no PRNG imports; no rendering; no plugin state.

Emitted rules artifact data/v3/rules/rules_artifact.jsonl (47 662 bytes, SHA e19fb205b282dabbf9f6ba38d97ed5364) 76 rules total across five rule types with the per-stem provenance schema:

Rule type	Count
harmonic	18
rhythmic	18
melodic	18
form	18
arrangement	4

Each rule row carries: `event_type`: "rule", `event_id` (16-hex content hash), `rule_id` (16-hex content hash), `rule_type`, `extractor` (extractor family, e.g. `extract.arrangement`), `extractor_version`

(v3-rules-c23-1), parameters (rule-type-specific typed body), parameters_random_state: 0, provenance_pointers (array pinning source song SHA-16, stem name, MIDI track index, PPQ, measure range), scope (level and time-window), schema_v: 1, confidence (deterministic per-rule score, e.g. 0.65 on arrangement rules with sensible window coverage), and ts (fixed 2026-09-02T00:00:00Z).

Sample rule (arrangement, first row on WIG):

```
{
  "rule_type": "arrangement",
  "extractor": "extract.arrangement",
  "extractor_version": "v3-rules-c23-1",
  "parameters": {
    "n_windows": 2,
    "windowBars": 4,
    "windows": [
      {
        "active_stems": ["bass", "drums", "piano", "vocals", "other"],
        "window_bar_start": 0
      },
      {
        "active_stems": ["piano", "vocals", "other"],
        "window_bar_start": 4
      }
    ]
  },
  "provenance_pointers": [
    {
      "song_sha16": "252eb21ce7df7328",
      "stem": "full_mix",
      "midi_track_index": -1,
      "midi_ticks_per_beat": 480,
      "measure_range": [0, 6]
    }
  ],
  "rule_id": "rule_5be3c9c61efabc51",
  "confidence": 0.65,
  "parameters_random_state": 0,
  "ts": "2026-09-02T00:00:00Z",
  "schema_v": 1
}
```

Self-anchor at data/v3/rules/rules_artifact.sha256 (65 bytes, SHA-16 25946f7d5b80874d) — a plain-text sidecar recording the artifact SHA for downstream cross-checking.

2.4 Byte-determinism ×2

The extractor was run twice into fresh `tempfile.mkdtemp()` directories under identical environment pins. Both runs produced the same artifact SHA `e19fb205b282dabbf9f6ba38d97ed53649d160a9bf36e9588e03b7c`. The verdict records this in `byte_determinism = { runs: 2, sha_equal: true, run_sha: "e19fb205..." }`. Byte-determinism ×2 gate PASS.

2.5 Anchor preservation

data/v3/rules/anchor_preservation_c23.json (5 588 bytes) records the read-only snapshot of 33 predecessor anchors, all matching pre-versus-post byte-identically. The three named read-only predecessor ledgers — c9 data/rules/ledger.jsonl, c15 ledger_i3_dminor.jsonl, c40 ledger_rated_corpus.jsonl — are all preserved unchanged (`all_match=true`). Writes to the fresh data/v3/rules/ shard did not touch any predecessor tree.

2.6 Test suite

tests/test_v3_rules_deterministic_extractor.py — 15 cases, 15/15 PASS. Cases cover: three-way rubric hash chain byte-equality; extractor imports without side effects; per-rule schema validation; per-rule content-hashed rule_id reproducibility; byte-determinism ×2 assertion (running the extractor twice and comparing SHAs); interpreter guard /usr/bin/python3 in the extractor's shebang; no PRNG imports (AST scan for random, numpy.random); no sidcar_nonfactor imports; no VST3 state APIs (AST scan for get_state, set_state, save_state); read-only preservation of the c9, c15, c40 predecessor ledgers; fetchability ladder shape; rules artifact non-empty; per-rule provenance-pointer non-empty; rule-type distribution matches the verdict's n_rules_by_type block; parameters_random_state=0 present on every rule.

2.7 Verdict

data/v3/rules/verdict.json (17-key schema) emitted with:

- milestone = M-V3-RULES-1
- cycle = 23, fork = d5530f8dlccc, clone = 2
- rubric_hash_v3_rules = e81ff589200f6d6b52d7a68f813e8120a544bfb371f8e0a45a2ac7f8793d71e5 (three-way chain byte-equal at document SHA == rubric_hash.txt content == verdict field)
- corpus_song_sha16s = [31a164f845f8e27e, 252eb21ce7df7328, 51e433ade2a845e1, cdd2717e52820ff6]
- n_rules_total = 76; n_rules_by_type = { harmonic: 18, rhythmic: 18, melodic: 18, form: 18, arrangement: 4 }
- rules_artifact_sha256 = e19fb205b282dabbf9f6ba38d97ed53649d160a9bf36e9588e03b7cd71ac8186
- byte_determinism = { runs: 2, sha_equal: true, run_sha: "e19fb205..." }
- tests_pass_over_total = [15, 15]
- extractor_version = v3-rules-c23-1; extract_ts_iso = 2026-09-02T00:00:00Z
- operator_ear_dependency = false (M-V3-RULES-1 has no ear-gate); panel_is_never_lands_gate = true
- env_pins = { PYTHONHASHSEED: 0, SOURCE_DATE_EPOCH: 1756463424, TZ: UTC, LC_ALL: C.UTF-8 }
- notes array: first activation of rules-hashed contract for M-V3 arc; extends c9 rule schema with per-stem provenance from v3 doctrine; no PRNG present, parameters_random_state=0 recorded per rule; panel is never a LANDS gate (FD-6); byte-determinism failure would be operator-decides per FD-1.

The verdict schema deliberately omits a verdict string field. This is disclosed honestly as MODERATE #2 for c24 handoff — the auditor's internal verdict of V3_RULES_LANDS_pending_operator (accepted at the prior turn) is recorded in the merge-report narrative, and downstream reporting can materialize a verdict label from the schema in a future cycle.

2.8 Housekeeping ledger

Eight events emitted to the shadow ledger data/v3/rules/ledger_c23_clone_2.jsonl (3 055 bytes): six named substantive M-V3-RULES-1/* rows plus two housekeeping. The recurring shadow-ledger drift pattern (MINOR-1 across the c20/c21/c22 fanout arcs) reproduces here: rows land in the shadow ledger and await concat into the primary promise_ledger.jsonl at post-merge integration per the c33/c48 auto-suffix concat path.

2.9 Required output artifact

docs/v3_rules_deterministic_extractor_spec_c23.md (13 058 bytes, SHA e81ff589200f6d6b52d7a68f813e8120a544bfb37) landed under docs/ per the directive. Merge report at workspace-root fallback path merge_

report_c23_clone_2_fork_d5530f8d1ccc.md (6 542 bytes) for root-conductor cp to the intended fanout path /home/user/music-gen-instance-v3/fork-d5530f8d1ccc/clone-2/merge_report.md at merge time (same c20/c21 clone-2 precedent for the workspace-sandbox path relocation).

3. Cycle 2: bookkeeping-only no-op and branch closure

Cycle 2's research brief was an explicit bookkeeping-only no-op mandate: zero writes, zero anchor touches, zero manufactured scope; the substantive work of Cycle 1 is complete and further activity would violate the no-null-cycle-validation rule and the anti-Hold-Pattern invariant. The worker honored the mandate verbatim: zero writes performed; zero new files under scripts/v3_rules/; zero writes under data/v3/rules/; zero tests/test_v3_rules_* edits; zero new ledger events.

The Cycle 2 auditor performed live SHA re-verification against every anchor from the Cycle 1 delivery:

Anchor	Live SHA-16	Byte-match
docs/v3_rules_ deterministic_extractor_ spec_c23.md	e81ff589200f6d6b (13 058 B)	☐
data/v3/rules/rubric_ hash.txt content	e81ff58920 0f6d6b...3d71e5	☐
data/v3/rules/verdict.json. hash_v3_rules	e81ff58920 0f6d6b...3d71e5	☐
data/v3/rules/rules_ artifact.jsonl	e19fb205b282dabb (47 662 B)	☐
data/v3/rules/rules_ artifact.sha256	25946f7d5b80874d (65 B)	☐ present
data/v3/rules/ledger_ c23_clone_2.jsonl	3 055 B on disk	☐ present
data/v3/rules/fetchability_ ladder.jsonl	484 B on disk	☐ present
data/v3/rules/anchor_ preservation_c23.json	5 588 B on disk	☐ present
merge_report_c23_clone_ 2_fork_d5530f8d1ccc.md	6 542 B on disk	☐ present

Three-way rubric_hash_v3_rules chain byte-equal at e81ff589200f6d6b52d7a68f813e8120a544bfb371f8e0a45a2ac7f87. Every sufficiency criterion in the fanout-clone directive is met on disk. Every anchor is unchanged. The clone's substantive scope is genuinely exhausted.

Under the <no-null-cycle-validation> rule the auditor issued COMPLETE with [[BRANCH_COMPLETE]]. The three MODERATE and one MINOR c24 handoff items (shadow-ledger schema rename, verdict.json label materialization, plain-assert test-convention policy call, main-ledger concat gated on schema rename) are explicit root-conductor / post-merge territory by construction, out of clone-2 scope.

4. Merge disposition and campaign-level significance

Merge disposition. This branch merges as [[BRANCH_COMPLETE]]. The required output artifact exists at the required path; the extractor prototype, rules artifact, self-anchor, 33-anchor preservation snapshot, 15/15 test suite, byte-determinism ×2 proof, eight-row shadow-ledger

shard, and workspace-fallback merge report are all on disk unchanged. Every hard anchor byte-identical pre-versus-post live-verified across both cycles.

Handoffs for root conductor c24+ (bookkeeping only; substantive on-disk artifacts already landed):

1. **Shadow-ledger main-concat (MODERATE, recurring non-blocking).** Eight-row shadow-ledger shard (six substantive M-V3-RULES-1/* plus two housekeeping) at `data/v3/rules/ledger_c23_clone_2.jsonl` awaits concat into primary `promise_ledger.jsonl` at post-merge integration per the c33/c48 auto-suffix concat path. Gated on the schema-rename item below.
2. **Shadow-ledger schema rename (MODERATE).** The row schema in the shadow ledger uses field names that need reconciliation with the primary ledger schema before concat. Root-conductor post-merge integration step.
3. **verdict.json label materialization (MODERATE).** The verdict's 17-key schema deliberately omits a verdict string field; the auditor's internal `V3_RULES_LANDS_pending_operator` is recorded in the merge-report narrative. Downstream reporting may want to materialize a verdict label field from the schema in a future cycle for consistency with c4-c22 verdict shapes.
4. **Plain-assert test convention (MINOR, campaign-wide policy call).** The c23 test suite uses plain assert statements rather than a pytest framework. Neither convention is wrong; the choice is a policy call the root conductor may want to make explicit for consistency across future rule/generation tests.

Campaign-level significance. This is the **first activation of the operator's rules-hashed contract for the M-V3 arc**. The rules-extraction step that feeds the eventual M-V3-GEN-1 generator is now instantiated, byte-deterministic under fixed environment pins, and reproducing across two independent runs at the same content-hashed SHAs. The 76-rule artifact is the campaign's first materialized bridge from the operator-approved reconstruction corpus (four v3 focus songs) to the downstream generation-half milestones. Every rule carries per-stem provenance pointing back to a specific song, stem, MIDI track index, and measure range, so a subsequent M-V3-GEN-1 that samples rules from this artifact can trace each generation-step's origin back to the operator-approved source material.

The extractor's implementation choice — pure stdlib plus mido, no music21/mingus/jsonschema/sklearn — is a Fixed Decision 1 win: dependency surface minimized, determinism guaranteed by canonical JSON plus fixed timestamps plus content-hashed rule IDs, no PRNG anywhere, no plugin state, no rendering. The fetchability ladder honestly records that the survey-open-source-first rule was applied and the on-disk availability of the four candidate library families was probed without any fetch attempt.

5. Campaign-level state at end of arc

M-V3-SPINE-1: operator-ear-LANDED since 2026-09-02 on Cycle 5 Chicken Grease reconstruction. Unchanged.

M-V3-FOCUS-1: closed with three-of-three operator-ear accepts (Chicken Grease mandatory + WIG + Disco A) on 2026-09-02. Rome and Peach Dream remain pending operator ear but non-blocking under the operator's "KEEP MOVING" directive.

M-V3-CORPUS-1: opening / status held by other fork-d5530f8d1ccc clones (out of this clone's scope).

M-V3-RULES-1: first activation LANDED at c23 clone-2 (this branch) with rules artifact `e19fb205b282dabb...`, three-way rubric chain byte-equal at `e81ff589200f6d6b...`, byte-determinism $\times 2$

PASS, 15/15 tests, 33 anchors preserved. No ear-gate.

M-V3-EAR-1, M-V3-GEN-1: downstream of M-V3-RULES-1; opening timing depends on root-conductor c24+ integration and any peer-clone parallel work in fork d5530f8d1ccc.

Discipline observations. Fifteen consecutive audits across this session with zero fabrications. Live-verification pattern (inline python3+hashlib SHA checks) unchanged. The recurring MINOR-1 shadow-ledger main-concat drift and MINOR-2 brief rubric-drift patterns are stable campaign-wide (present on c20, c21, c22, c23); workers correctly adapt to on-disk truth per FD-1; neither blocks LANDS when substantive artifacts are on disk. All banned anti-patterns (VST3 state extraction under c31 STILL_GAP + c35 SPINE, CLAP HF SSL fetch under c11, M-EAR-1 Path A audits under N=55, c37 pretty_midi merge_partial) had zero re-attempts this cycle.

6. Conclusions

Clone 2 of fork d5530f8d1ccc executed the c23 first-activation of M-V3-RULES-1 cleanly under the operator's determinism-for-generation-half stance and the standing survey-open-source-first rule. The rubric document was pinned before any script was written; the extractor prototype was implemented as pure stdlib plus mido (no external dependency surface beyond what the campaign already vendors); the four operator-approved v3 focus-song merged.mid deliveries were consumed read-only; 76 typed rules landed across five rule types with per-stem provenance in the extended c9 schema; byte-determinism $\times 2$ held across two fresh-tempdir runs at the same content-hashed SHA; 15 of 15 tests passed; the three-way rubric_hash_v3_rules chain held byte-equal; every read-only predecessor ledger (c9, c15, c40) preserved unchanged. Cycle 2 was a correct bookkeeping-only no-op; the branch closes under [[BRANCH_COMPLETE]].

This is the campaign's first materialization of a rules-hashed contract inside the M-V3 arc. The 76-rule artifact becomes the input a subsequent M-V3-GEN-1 milestone will consume, with each rule traceable back to a specific operator-approved song, stem, and measure range. The generation-half of the campaign is now open in a form that carries the same anti-fabrication and byte-determinism disciplines that carried the reconstruction-half from Cycle 3 forward.

Appendix: Implementation Details

A.1 Delivered artifacts

Required output artifact: docs/v3_rules_deterministic_extractor_spec_c23.md (13 058 bytes, SHA e81ff589200f6d6b52d7a68f813e8120a544bfb371f8e0a45a2ac7f8793d71e5).

Rules artifact: data/v3/rules/rules_artifact.jsonl (47 662 bytes, SHA e19fb205b282dabbf9f6ba38d97ed53649d160a9b).

Self-anchor: data/v3/rules/rules_artifact.sha256 (65 bytes, SHA-16 25946f7d5b80874d).

Pinned rubric hash file: data/v3/rules/rubric_hash.txt (content e81ff589200f6d6b52d7a68f813e8120a544bfb371f8e0a).

Verdict: data/v3/rules/verdict.json (17-key schema, three-way chain byte-equal).

Fetchability ladder: data/v3/rules/fetchability_ladder.jsonl (484 bytes).

Anchor preservation snapshot: data/v3/rules/anchor_preservation_c23.json (5 588 bytes, 33 anchors, all_match=true).

Shadow ledger: data/v3/rules/ledger_c23_clone_2.jsonl (3 055 bytes, 8 events).

Extractor: scripts/v3_rules/extract_rules.py.

Test suite: tests/test_v3_rules_deterministic_extractor.py (15 cases, 15/15 PASS).

Merge report workspace-fallback: merge_report_c23_clone_2_fork_d5530f8d1ccc.md (6 542 bytes);
root-conductor cp to /home/user/music-gen-instance-v3/fork-d5530f8d1ccc/clone-2/merge_report.md
at merge time.

A.2 Integrity chains

Three-way rubric_hash_v3_rules chain: docs/v3_rules_deterministic_extractor_spec_c23.md SHA
e81ff589200f6d6b52d7a68f813e8120a544bfb371f8e0a45a2ac7f8793d71e5 == data/v3/rules/rubric_
hash.txt content == verdict rubric_hash_v3_rules field.

Rules-artifact self-anchor: data/v3/rules/rules_artifact.sha256 content == data/v3/rules/rules_
artifact.jsonl SHA-256 == verdict.rules_artifact_sha256 == e19fb205b282dabbf9f6ba38d97ed53649d160a9bf36e9588

A.3 Corpus consumed (READ-ONLY)

Four operator-approved v3-rendered focus-song deliveries via their merged.mid:

Song	sha16	Source cycle
Chicken Grease	31a164f845f8e27e	c5 (operator-blessed 2026-09-02)
What If I Go	252eb21ce7df7328	c21 clone-1 restart (merged.mid SHA a93f5c2ae16e5cace42b98886f6ce3eae4bb47393bef9
Rome	51e433ade2a845e1	c20 clone-1 (merged.mid SHA c28b8686684fddfc841a27e96e299a93f1099fe99a5de
Disco A	cdd2717e52820ff6	c21 clone-0 (merged.mid SHA-16 7e6f131f07f0d33c)

A.4 Rules artifact composition

76 rules total across 5 rule types:

Rule type	Count	Extractor family
harmonic	18	extract.harmonic
rhythmic	18	extract.rhythmic
melodic	18	extract.melodic
form	18	extract.form
arrangement	4	extract.arrangement

Per-rule schema fields: event_type, event_id (16-hex), rule_id (16-hex content-hashed), rule_
type, extractor, extractor_version (v3-rules-c23-1), parameters (typed body), parameters_random_
state: 0, provenance_pointers (song SHA-16, stem, MIDI track index, PPQ, measure range),
scope (level, start_s, end_s), schema_v: 1, confidence (deterministic per-rule score), ts (fixed
2026-09-02T00:00:00Z).

A.5 Byte-determinism ×2 proof

`byte_determinism = { runs: 2, sha_equal: true, run_sha: "e19fb205b282dabbf9f6ba38d97ed53649d160a9bf36e9588e03b7cc" }` recorded in the verdict. Two fresh `tempfile.mkdtemp()` runs under identical env pins produced byte-identical artifacts.

A.6 Test suite (15/15 PASS)

Cases: three-way rubric chain byte-equality; extractor imports without side effects; per-rule schema validation; per-rule content-hashed `rule_id` reproducibility; byte-determinism ×2 assertion; interpreter guard `/usr/bin/python3`; no PRNG (AST scan for `random`, `numpy.random`); no `sidecar_nonfactor`; no VST3 state APIs (AST scan for `get_state`, `set_state`, `save_state`); read-only preservation of c9/c15/c40 predecessor ledgers; fetchability ladder shape; rules artifact non-empty; per-rule provenance-pointer non-empty; rule-type distribution matches `n_rules_by_type`; `parameters_random_state=0` on every rule.

A.7 Fetchability ladder (survey-open-source-first)

Four candidate library families probed on-disk without any fetch attempts:

- `music21` — on-disk availability recorded in `fetchability_ladder.jsonl`.
- `mingus` — on-disk availability recorded.
- `jsonschema-driven grammar` — recorded.
- `sklearn` — recorded.

Implementation decision per the ladder: pure `stdlib` + `mido` (already vendored via campaign's `mido 1.3.3` pin).

A.8 Anchor preservation (33 anchors byte-identical pre==post)

Recorded at `data/v3/rules/anchor_preservation_c23.json` with `all_match=true`. Includes the three named predecessor ledgers: c9 `data/rules/ledger.jsonl`; c15 `ledger_i3_dminor.jsonl`; c40 `ledger_rated_corpus.jsonl` — all byte-identical.

A.9 Verdict schema (17 keys, verdict label deliberately omitted)

Keys: `byte_determinism`, `clone`, `corpus_song_sha16s`, `cycle`, `env_pins`, `extract_ts_iso`, `extractor_version`, `fork`, `milestone`, `n_rules_by_type`, `n_rules_total`, `notes`, `operator_ear_dependency`, `panel_is_never_lands_gate`, `rubric_hash_v3_rules`, `rules_artifact_sha256`, `tests_pass_over_total`.

Auditor internal verdict: `V3_RULES_LANDS_pending_operator` (recorded in merge-report narrative; MODERATE #2 handoff for materialization).

A.10 Environment pins

`PYTHONHASHSEED=0`; `SOURCE_DATE_EPOCH=1756463424`; `TZ=UTC`; `LC_ALL=C.UTF-8`; `OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`, `OPENBLAS_NUM_THREADS=1`; interpreter `/usr/bin/python3`; `mido==1.3.3`. Recorded in `verdict.env_pins`.

A.11 Handoffs for root conductor c24+

MODERATE #1 (shadow-ledger main-concat, recurring non-blocking): 8-row shadow shard at `data/v3/rules/ledger_c23_clone_2.jsonl` awaits c33/c48 auto-suffix concat.

MODERATE #2 (schema rename): shadow-ledger row schema needs reconciliation with primary ledger schema before concat.

MODERATE #3 (verdict.json label materialization): 17-key schema omits verdict string field; auditor's internal V3_RULES_LANDS_pending_operator recorded in merge-report narrative; downstream may materialize label in future cycle for consistency with c4-c22 verdict shapes.

MINOR (plain-assert test convention, campaign-wide policy call): c23 test suite uses plain assert rather than pytest framework; policy call for consistency across future rule/generation tests.

A.12 Source sessions

Cycle	Researcher	Worker	Auditor
1	00e31c13-4488-44a0-afe8-4d7537ff24ba	e02570aa-52a3-4ebc-be9c-8a145cf021d5	3dba5d81-7879-4414-b9ad-07774442fefafa
2	c812527b-7cfe-49ef-87a8-b6714950edfa	9eb6eae1-39b3-47b1-b54b-39a69952f8b8	ab6a0e07-cc65-4a46-ad8e-d730fd2e599c

A.13 Fanout metadata

Fork d5530f8d1ccc. Clone 2 of the M-V3-RULES-1 first-activation assignment. Merge report expected at /home/user/music-gen-instance-v3/fork-d5530f8d1ccc/clone-2/merge_report.md for root-conductor pickup; workspace-root fallback at merge_report_c23_clone_2_fork_d5530f8d1ccc.md. Sibling clones (parallel opening work on other downstream milestones) reported separately.